

# Hou's Navier-Stokes singularity scenario: a resolved front in the class excluded by the Liouville theorems, and its certified centrifugal instability

[Author]  
[Affiliation]

*Draft 0.9, 2026-09-08 (late-stage transition at  $\nu = 5 \times 10^{-4}$  added to 5.2, 6.0 and 7.5). All numbers final at this stage. Author list, affiliations and acknowledgements to be added.*

## Abstract

Hou's interior scenario for singularity formation in the three-dimensional axisymmetric Euler equations, and its Navier-Stokes continuation with a two-stage viscosity, are the most developed candidates for a finite-time singularity of a smooth finite-energy incompressible flow. We continue the viscous scenario at Hou's first-stage viscosity to 331 times the initial amplitude with a front-following axial map that resolves the sharp front to ten cells (six at the last point), and measure the quantity  $\sup |u| |x'|$  that is the hypothesis of the Koch-Nadirashvili-Seregin-Sverak Liouville theorem for bounded ancient axisymmetric solutions with swirl. It does not involve the fitted singular time and is constant to 0.3 percent (72.2) from 144 to 331 times the initial amplitude, over 3.3 units of  $\log(1/(T - t))$ ; its value is carried by the outer flow, and at the swirl maximum itself the same quantity is flat at 21 to 22 where a logarithmic growth would read thirty percent. Over the resolved range the scenario as computed therefore satisfies the hypothesis under which the exclusion theorems apply, which is the opposite of the logarithmic escape Hou identified in his generalized model at far larger amplifications; a finite computation cannot establish the bound up to  $T$ , and on the core-restricted  $ru^r$  the data are inconclusive (non-monotone between 15 and 22 over the range). The front is the Rott-Lundgren strained layer under the measured collapse strain to a five percent profile residual, the amplitude is Type I (exponent  $1.03 \pm 0.03$ ) with a converged singular time ( $0.0022865 \pm 1 \times 10^{-7}$ ) by a log-derivative fit, the meridional Type I constant is bounded, and the scale-invariant local  $L^3$  norm at the collapse scale decreases over four units of  $\log(1/(T - t))$ . We also compute, for the first time, the linear stability of the interior collapse to non-axisymmetric perturbations, on the frozen and on the co-evolving base, in Euler and in Navier-Stokes at both of Hou's viscosities, with resolution certificates in both grid directions: the long-wave axisymmetric direction is marginally stable (peak 0.495 of the collapse rate at three resolutions), while the short-wave shearing continuum of the front is unstable relative to the collapse at every azimuthal wavenumber tested ( $m = 0, 1, 2$ ), crossing half the collapse rate at 9 to 12 percent of the remaining time at viscosity  $5 \times 10^{-4}$  and at 27 to 37 percent at  $5 \times 10^{-3}$ ; the crossing fraction of the  $m = 1$  family rises roughly logarithmically with viscosity from  $5 \times 10^{-4}$  to  $1 \times 10^{-2}$ . Nonlinearly, every seed, state and viscosity tested depletes the collapse rather than transferring it. The instability is identified as the centrifugal instability of the front, whose circulation decreases outward, extended to every azimuthal wavenumber by the axial flow of the collapse: a one-dimensional parallel-flow model of the front column has an isolated unstable eigenvalue whose inviscid rate approaches the local Rayleigh bound (0.96 of the collapse rate) from below and whose viscous peak (0.81) brackets the two-dimensional frozen-base rate (0.80; the model is local, its rate spans 0.5 to 1.05 across the front's thickness and it fails on the inviscid state), and that eigenvalue is certified by a Krawczyk enclosure with rigorous floating-point bounds for the collocation discretization of the

model on an explicit rational polynomial field, at three resolutions, the two finest agreeing to nine digits; the Leibovich-Stewartson sufficient condition is certified on the same kind of field by Sturm root counting in exact arithmetic. The local front structure is an exact infinite-energy Navier-Stokes solution, which explains why a core-resolving computation cannot observe the departure that finite energy forces at a few collapse lengths.

## 1 Introduction

The regularity problem for the three-dimensional incompressible Navier-Stokes equations remains open in both directions. On the numerical side the strongest evidence for singular behaviour comes from the axisymmetric flows with swirl of Luo and Hou [LH14a, LH14b] at a solid boundary and, in the interior, of Hou [H23a] for Euler and Hou [H23b] for Navier-Stokes. The boundary scenario now has a computer-assisted proof of stable nearly self-similar blowup for Euler [CH25]; the interior scenario is supported by a self-similar profile fitted through amplifications of order  $10^7$  in Euler and, with a viscosity of  $5 \times 10^{-4}$  raised to  $5 \times 10^{-3}$  at a chosen time, in Navier-Stokes.

Two questions about these scenarios have not been asked in print. First, every stability statement about them, Hou’s data-perturbation tests, the nonlinear stability of Chen and Hou [CH25], and the unstable-direction counts of the machine-learned profiles of Wang and coauthors [WLGB23, W25], is a statement within the axisymmetric class; whether the collapsing state is stable to symmetry-breaking perturbations is unknown. Second, the viscous scenario’s relation to the exclusion theorems for Type I singularities of axisymmetric flows with swirl [CSTY09, KNSS09, SS09] has been argued about but not measured: Hou identifies a logarithmic growth of  $ru^r$  as the mechanism by which his solution escapes those theorems [H24], and Barker [B25] has responded that the theorems’ hypotheses do not involve that quantity.

This paper answers the first question and measures the quantities relevant to the second. Section 2 describes the codes and their validation. Section 3 gives the spectral table: the growth rates of azimuthal modes  $m = 0, 1, 2$  about the collapsing state, in units of the collapse rate, with resolution certificates in both grid directions, together with the dependence of the  $m = 1$  crossing on viscosity, the identification of the instability as the centrifugal instability of the front column, and a computer-assisted certificate of its discrete eigenvalue. Section 4 gives the nonlinear fate of the unstable perturbations. Section 5 gives the viscous continuation: the singular-time fit, the front-following map, and the identification of the front. Section 6 places the measured quantities against the theorems. Section 7 discusses what the results mean for constructions of Type II singularities and for the regularity side of the problem.

The results are negative in the sense the singularity literature cares about: the leading interior scenario is unstable to symmetry breaking, the instability depletes rather than intensifies the collapse, and the viscous continuation sits inside the class the theorems exclude. They are positive as characterization: the spectral table is the first non-axisymmetric stability computation of an axisymmetric blowup candidate, the front is identified as an exact local solution, and the theorem gap is stated with the measured constants attached.

## 2 Equations, codes and validation

### 2.1 Formulation

We use the Hou-Luo variables  $u_1 = u_\theta/r$ ,  $\omega_1 = \omega_\theta/r$ ,  $\psi_1 = \psi/r$  on the periodic cylinder  $r \leq 1$ ,  $z \in [0, 1)$  with the axisymmetric Navier-Stokes equations in the form of [H23a, H23b], the same initial data (the swirl profile  $12000(1 - r^2)^{18} \sin(2\pi z)/(1 + 12.5 \sin^2(\pi z))$ ), and the same boundary

condition at the wall. The base solver (axiphys) integrates these equations on a radial sinh-type map and an axial map, with the Hou-Li exponential filter, a moving radial map that follows the collapse, and, for the runs of Section 5, a front-following axial map described there. It reproduces Hou’s published checkpoints: the Euler vorticity ratio at his first checkpoint to one percent and the Navier-Stokes ratio at  $t = 0.0022616$  to three percent (x144 against his 139.6); the details of the cross-code comparison, including the spread among Hou’s own tabulated and fitted values, are in the supplementary record (Rounds 86-87).

## 2.2 Azimuthal-mode solvers

Perturbations of azimuthal wavenumber  $m$  about an axisymmetric base are computed in three ways. The frozen-base solver (axi3dlin) linearizes about a snapshot and integrates the mode with an exact discrete pressure projection (the divergence-of-gradient operator assembled from the same difference stencils, banded, with a Neumann wall row so that the projected radial velocity vanishes at the wall) and the parity rules  $u'_r, u'_\theta \sim r^{m-1}$ ,  $u'_z, p' \sim r^m$  at the axis; viscosity enters through the combinations  $u_r \pm iu_\theta$ , which obey the Laplacians of index  $m \pm 1$ . The tangent-linear solver (axi3dttl) marches the nonlinear axisymmetric base and the mode in one Runge-Kutta step, linearizing about the instantaneous base at every stage, with the base’s moving radial map installed as a fixed map and mapped bases transferred by exact sine series. The nonlinear solver (axi3dnl) carries modes  $m = 1$  to  $M - 1$  with the azimuthal convolutions evaluated pseudo-spectrally on  $3M$  points and feeds the  $m = 0$  Reynolds stress back into the base equations.

For  $m = 0$  the seed exponents become  $u'_r, u'_\theta \sim r$  and  $u'_z \sim 1$ , and the axial-mean Neumann problem for the pressure is singular and is pinned at the axis.

## 2.3 Gates

Solid-body rotation is neutral to  $1 \times 10^{-10}$  in the frozen solver. A Rayleigh-unstable column reproduces a one-dimensional primitive-variable eigensolver (0.3708 for  $m = 0$ ;  $0.248 + 0.337i$  for the physical  $m = 1$  mode, with wall-localized spurious modes identified and excluded). The nonlinear code with a seed of  $1 \times 10^{-6}$  reproduces the tangent-linear rate to three digits (0.158 against 0.160) and generates the  $m = 2$  harmonic at second order. The tangent-linear rate at equal fraction of the remaining time reproduces the frozen-base rate to two percent up to a fifth of the remaining time.

## 2.4 The trivial directions and the threshold

On a Type I base  $u = (T - t)^{-1/2}U(x/\sqrt{T - t})$  the amplitude rate  $\sigma$  of a perturbation, in units of the collapse rate  $1/(T - t)$ , is  $3/2$  for the time-translation direction  $d_t U$  and 1 for an axis translation; a perturbation outruns the collapse (its ratio to the base grows) when  $\sigma(T - t) > 1/2$ . For  $m = 1$  the axis-translation field  $-d_x U$  is an exact tangent-linear solution and is projected out with the energy inner product (exact for a normal operator; the overlap stays below 0.3 in all runs, so the projection error is at the ten percent level or below); for  $m = 0$  the overlap with  $d_t U$  is monitored and stays below 0.1. We report both the raw rate and the rate of the orthogonal part where it matters.

## 2.5 Certificates

Every long-wave entry of the table has three axial resolutions (512, 1024 and 2048 points, with the localized seed pinned at the same position) and a radial pair (an exact subsample of the sinh-map grid to 257 points against 513). The short-wave families have an axial pair (1024 against 2048)

on the viscous base. The viscous continuation has a twin at a different axial map and Courant number, and the front-resolved runs have two maps. Three earlier readings were retracted during the work and are recorded as such in the supplementary log: a fit-dependent ‘departure’ of the Type I constants (the singular-time degeneracy of Section 5.1), a grid-scale event in a seeded run at the x200 state, and a three-cell front thickness. We regard the retractions as part of the certificate.

### 3 The spectral table

#### 3.1 Euler, the x28 state

The base is the Euler solution at  $t = 0.002208$ , where the swirl amplitude is 28 times its initial value, the core sits at  $(R, Z) = (0.0250, 0.0078)$ , and the Type I constant  $\sup |u| (|x| + \sqrt{T - t})$  is 75 (it is 75 on every state from x28 to x200, set by the outer flow).

*Long-wave  $m = 1$*  (Figure effig:1 collects the Euler rates of this section; Figure effig:5 the resolution pairs). On the frozen base the localized seed grows at 0.184, 0.232, 0.275, 0.313, 0.346, 0.375, 0.401, 0.425, 0.448 at 2.2, 4.3, 6.5, 8.7, 10.9, 13.0, 15.2, 17.4, 19.5 percent of the remaining time; at three axial resolutions the rate at 8.7 percent reads 0.318, 0.313, 0.319 and at 13 percent 0.379, 0.375, 0.380 (512, 1024, 2048 points: agreement to 1.3 percent), and the radial subsample reads 0.306 at 8.7 and 0.351 at 11.6 percent (2 percent). On the co-evolving base the rate rises 0.25, 0.34, 0.42, 0.49, 0.56, 0.64, 0.71, 0.76, 0.82 at 6, 11, 17, 23, 29, 34, 40, 45.6, 46.3 percent, thirteen percent above the frozen rate at equal fraction; after projecting out the translation direction the orthogonal part reads 0.40, 0.55, 0.62, 0.67, 0.80 at 23, 30, 38, 44, 47 percent. The non-trivial  $m = 1$  perturbation therefore outruns the Euler collapse from about 30 percent of the remaining time and reaches 0.8 of the collapse rate by 47 percent. Its axial wavenumber centroid co-scales with the collapse (7 to 30 cycles per unit as the core height shrinks), the swirl share of its energy production rises from 0.07 to 0.39, and its precession rate  $\omega(T - t)$  passes from +0.12 through zero to retrograde near 40 percent; the swirling base has only  $SO(2)$  symmetry, so this reversal is permitted and is not a bifurcation signature.

*Long-wave  $m = 2$* . 0.31 at 17 percent on the frozen base, rising.

*Long-wave  $m = 0$* . The localized seed on the frozen base rises to a maximum of 0.495 at 11.7 percent and falls to 0.378 by 29 percent; at three axial resolutions the maximum reads 0.495, 0.496, 0.497 (512, 1024, 2048 points) and the radial pair agrees to 1 percent (0.484 against 0.489 at 8.7 percent, 0.497 against 0.495 at 11.6). On the co-evolving base it plateaus at 0.454 at 10 percent and falls to 0.389 by 27 percent, with the overlap with  $d_t U$  at 0.08 to 0.10. The long-wave axisymmetric direction is stable: no non-trivial axisymmetric unstable direction exists at this level, consistent with Hou’s report that the profile is ‘very stable’ to perturbations of the data [H23b, Sec. 3.6].

*Short-wave families*. A smooth random seed under the same envelope excites, for every  $m$ , a family living in a thin band on the front whose axial wavenumber centroid marches with the collapse (17 to 80 cycles per unit for  $m = 0$ ; 26 to 54 for  $m = 1$  over the resolved range) and whose production is entirely from the meridional shear. Its rate has no maximum in Euler: for  $m = 1$  it reaches 0.9 of the collapse rate by 16 percent; for  $m = 0$  it crosses  $1/2$  at 8 percent and reaches 0.72 by 17 to 20 percent before the axial resolution limits it. This is the Kelvin-Helmholtz continuum of the front, in Hou’s words the ‘shearing instability’ that ‘typically occurs near the sharp front’ and that his numerical viscosity is chosen to suppress [H23a, Sec. 3.6].

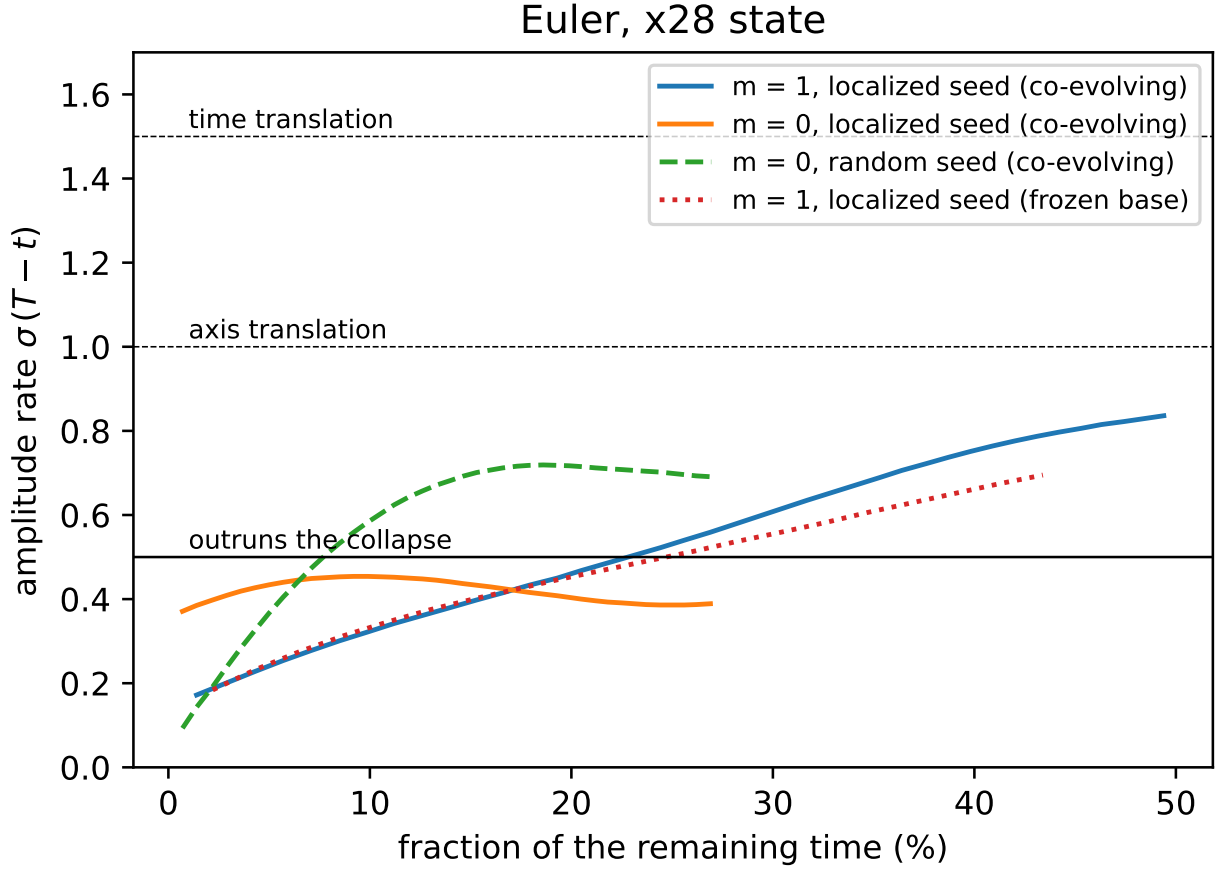


Figure 1: Euler, x28 state: amplitude rates  $\sigma(T-t)$  of the  $m = 1$  and  $m = 0$  localized seeds and the  $m = 0$  random seed on the co-evolving base (tangent-linear), and of the  $m = 1$  localized seed on the frozen base, against the fraction of the remaining time. The line at  $1/2$  is the threshold at which a perturbation outruns the collapse; the lines at  $1$  and  $3/2$  are the axis-translation and time-translation directions.

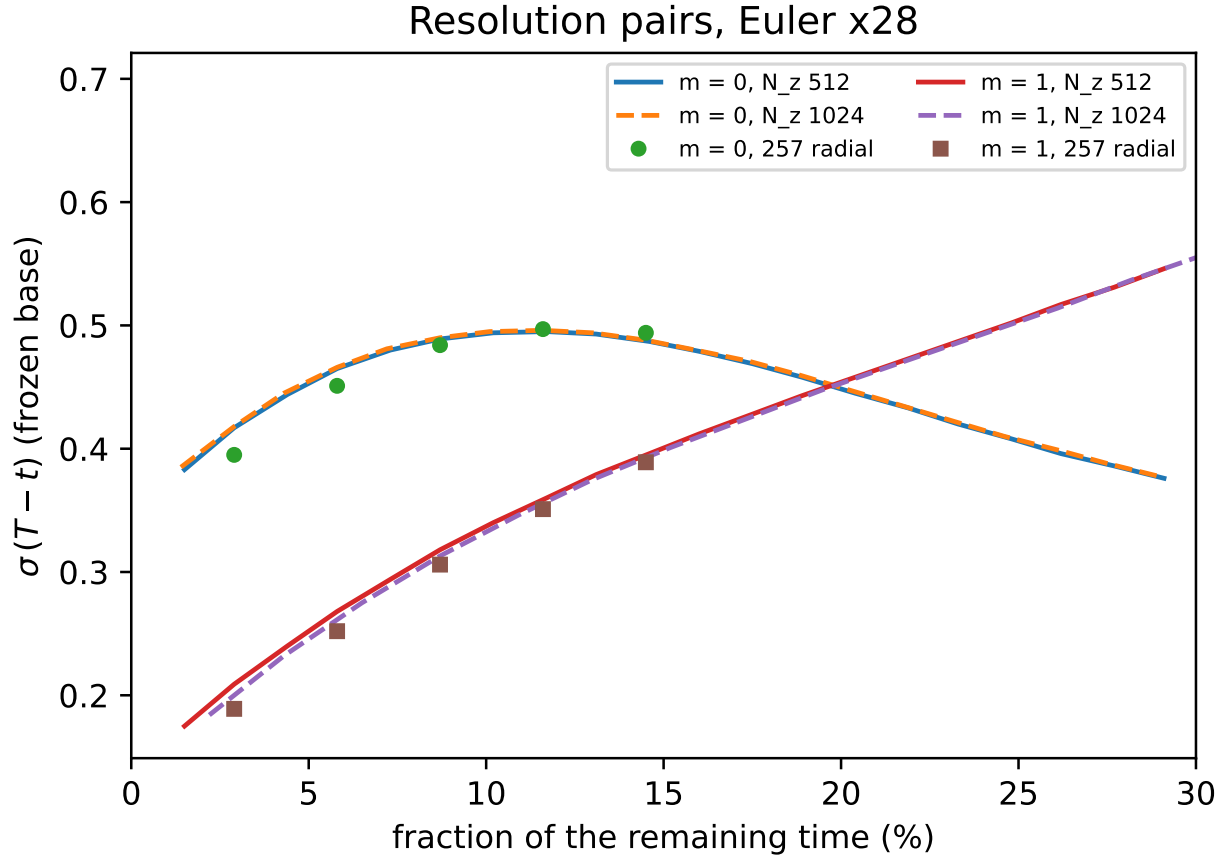


Figure 2: Resolution pairs on the Euler x28 state: frozen-base rates  $\sigma(T-t)$  of the long-wave  $m = 0$  and  $m = 1$  entries at 512 and 1024 axial points and on the 257-point radial subsample.

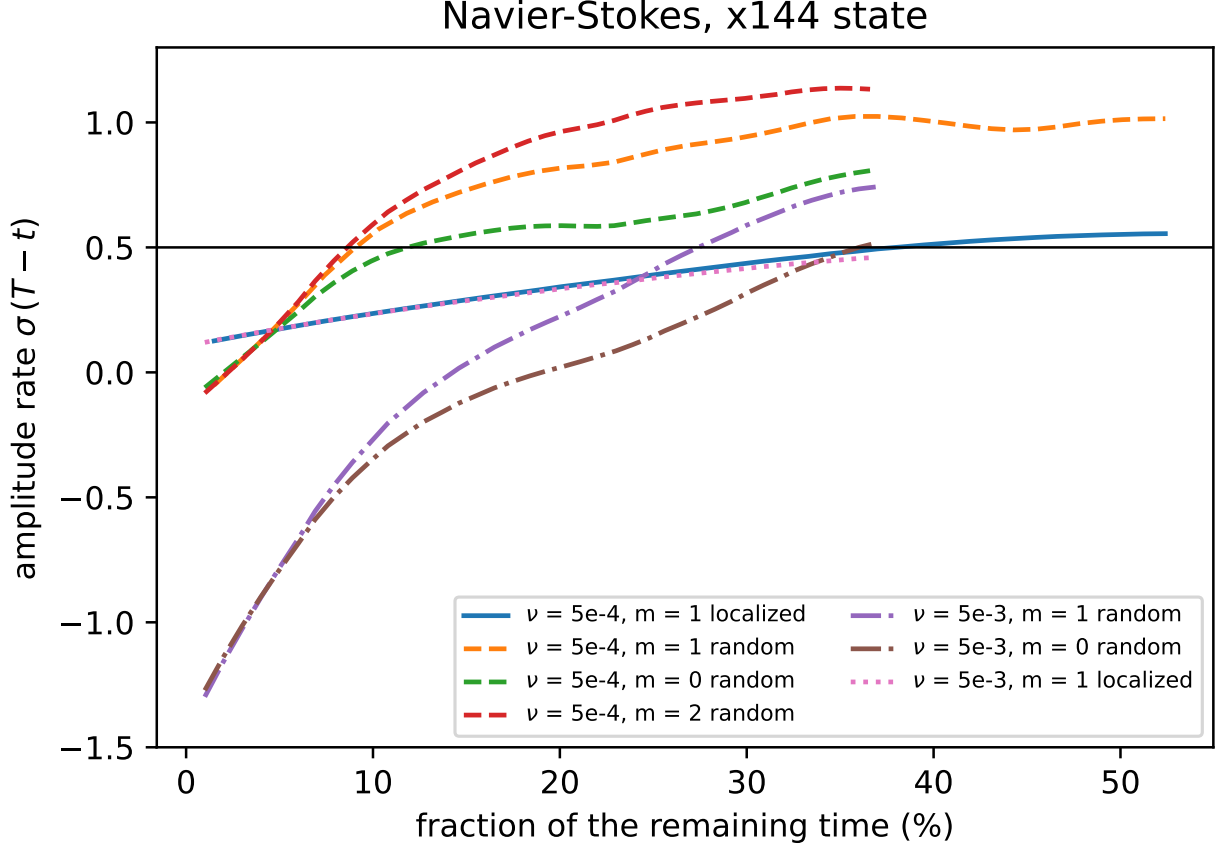


Figure 3: Navier-Stokes, x144 state: amplitude rates  $\sigma(T-t)$  on the co-evolving base at  $\nu = 5 \times 10^{-4}$  ( $m = 1$  localized seed;  $m = 0, 1, 2$  random seeds) and at  $\nu = 5 \times 10^{-3}$  ( $m = 0, 1$  random seeds;  $m = 1$  localized seed), against the fraction of the remaining time. The line at  $1/2$  is the threshold.

### 3.2 Navier-Stokes, the x144 state, viscosity $5 \times 10^{-4}$

The base is Hou's first-stage solution at  $t = 0.002262$  (amplitude 144 times the initial, his tabulated 139.6), whose amplitude still grows like  $(T-t)^{-0.85}$ .

*Long-wave  $m = 1$*  (co-evolving; Figure effig:2 collects the viscous rates of Sections 3.2 and 3.3): 0.13, 0.19, 0.24, 0.30, 0.37, 0.43, 0.48, 0.50 at 1.5, 6, 10, 16, 23, 29, 36, 38 percent; orthogonal part 0.32, 0.34, 0.39, 0.39 at 24, 32, 37, 40 percent, and 0.44 at 51 percent. Viscosity holds the long-wave mode below the threshold on this base.

*Short-wave families* (co-evolving, random seed):  $m = 1$  reads 0.475, 0.627, 0.768 at 10, 14, 18 percent, with the frozen-base axial pair agreeing to 3 to 7 percent (0.116 against 0.093 at 4 percent, 0.381 against 0.368 at 8);  $m = 0$  crosses  $1/2$  at 12 percent and reads 0.58 at 20 percent;  $m = 2$  crosses at 9 percent and saturates near 1.13 by 37 percent. The front's continuum is unstable relative to the collapse at every azimuthal wavenumber, with rates within about twenty percent of one another: it is a property of the sheet, not of a symmetry class. The sheet Reynolds number  $\Delta U/\sqrt{\nu a}$  with the measured velocity jump (Section 5) is 120 to 150 and, under the Type I scaling, constant in time, so the family is never frozen out.

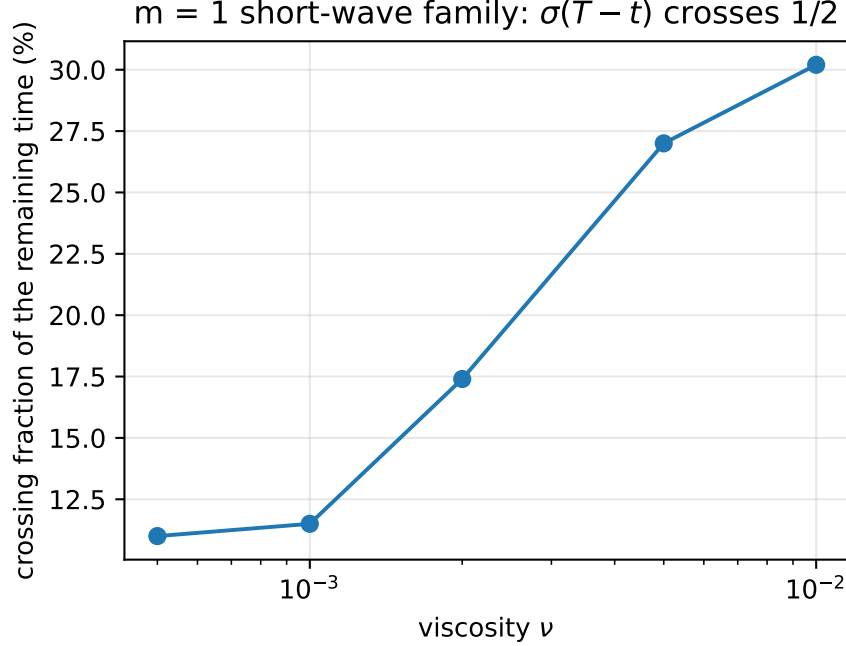


Figure 4: The  $m = 1$  short-wave family on the co-evolving x144 base: the fraction of the remaining time at which  $\sigma(T - t)$  crosses  $1/2$ , against the viscosity  $\nu$ .

### 3.3 Navier-Stokes, the x144 state, viscosity $5 \times 10^{-3}$

Hou raises the viscosity tenfold at  $t_0 = 0.00227375$  because  $5 \times 10^{-4}$  ‘is not strong enough to stabilize the shearing instability generated by the 3D Euler equations’ [H23b, Sec. 3]. Applying  $5 \times 10^{-3}$  to base and mode from the x144 state, a random seed first decays (rates  $-1.3$  in collapse units at one percent) and its surviving part re-grows:  $m = 1$  crosses zero at 14 percent and  $1/2$  at 27 percent and reads  $0.74$  at 37 percent;  $m = 0$  crosses zero at 19 percent and  $1/2$  at 37 percent; the long-wave  $m = 1$  direction rises slowly to  $0.46$  at 36 percent and stays below the threshold. The second-stage viscosity delays the axisymmetric family from 12 to 37 percent and the non-axisymmetric one from about 11 to 27 percent; it removes neither. The base at  $5 \times 10^{-3}$  lags Type I from the start (amplitude  $1.048$  against  $1.064$  at 6 percent).

### 3.4 The crossing fraction versus viscosity

For the  $m = 1$  random-seed family on the co-evolving x144 base, the fraction of the remaining time at which the rate crosses  $1/2$  (Figure 4) is 11 percent at  $5 \times 10^{-4}$ , 11.5 at  $1 \times 10^{-3}$ , 17.4 at  $2 \times 10^{-3}$ , 27 at  $5 \times 10^{-3}$  and 30.2 at  $1 \times 10^{-2}$  (the last run starting at  $-2.6$  and crossing zero at 20.5 percent). Over the decade from  $1 \times 10^{-3}$  to  $1 \times 10^{-2}$  the crossing fraction rises by nineteen points, roughly linearly in the logarithm of the viscosity and flattening at the top. At every viscosity from Hou’s first stage to twice his second, the non-axisymmetric family outruns the collapse before a third of the remaining time has elapsed; extrapolating the trend, holding it below the collapse rate to half the remaining time would need a viscosity of order  $3 \times 10^{-2}$ , at which the base no longer collapses like Type I.

### 3.5 An analytic cross-check

Leibovich and Stewartson’s sufficient condition [LS83] for non-axisymmetric short-wave instability of a swirling flow,  $2V\Omega [D(rV) D\Omega + D(W^2)] < 0$  with  $V$  the swirl,  $\Omega = V/r$ ,  $W$  the axial velocity and  $D$  the radial derivative, is evaluable on the axisymmetric base alone. It is satisfied in a thin annulus just outside the swirl maximum on every state examined: at  $(r, z) = (0.0273, 0.0078)$  against the maximum at  $(0.0250, 0.0078)$  on the Euler x28 state, at  $(0.0141, 0.0037)$  against  $(0.0138, 0.0037)$  on the viscous x144 state, and at  $(0.0030, 0.0009)$  against  $(0.0029, 0.0008)$  at x222; the implied short-wave growth is of order the collapse rate. Rayleigh’s centrifugal discriminant is negative on 6, 12 and 33 percent of the column through the swirl maximum inside twice its radius on the three states.

The evaluation at the viscous x144 state is made rigorous as a statement about an explicit field. Let  $\Omega(s)$  and  $W(s)$  be the degree-20 Chebyshev polynomials in  $s = (r/r_{\max})^2$ ,  $r_{\max} = 2R$ , with exactly rational coefficients (the double-precision least-squares fit to the column through the swirl maximum, which reproduces the grid column to 1.9 percent in the swirl and 4.5 percent in the axial velocity), and let  $(V, W) = (r\Omega, W)$  be the columnar vortex they define. Then  $\Phi = 2r\Omega^2 B(r)$  with  $B$  a polynomial of degree 80 with rational coefficients, and by Sturm’s theorem in exact rational arithmetic  $B$  has no real root in  $[0.013898, 0.015068]$  while  $B$  is negative at the midpoint; hence  $\Phi < 0$  on  $1.009R \leq r \leq 1.094R$ . The explicit columnar field therefore satisfies Leibovich and Stewartson’s sufficient condition [LS83] for non-axisymmetric short-wave instability on that annulus. (The polynomial reproduces the column’s values to a few percent but its first derivative is wrong by up to a factor two at the swirl maximum, which matters for eigenvalues and not for this sign certificate; see Section 3.6. The computation is in the accompanying script; an interval-arithmetic evaluation of the Chebyshev series failed by the dependency problem at this degree, which is why the exact polynomial route is used. The same certificate on the Euler x28 column, degree 30 on  $[0, 1.5R]$ , holds on  $0.998R \leq r \leq 1.197R$ , but there the polynomial reproduces the grid column only to 16 percent in the swirl because of the sharper inviscid front; we report it as supplementary.)

### 3.6 The columnar model of the front and a certified eigenvalue

*The columnar model.* A column of the flow at fixed height, treated as a parallel swirling flow  $(V, W) = (r\Omega, W)$  with axial velocity, has normal modes  $\exp(im\theta + ikz - i\omega t)$  governed by a one-dimensional eigenvalue problem, which we solve by Chebyshev collocation with the axis conditions imposed by parity and no-slip at the outer edge of the fitted column ( $2R$ ). Two cautions attach to the certificate of Section 3.5. The first concerns the field: the certificate is a statement about the explicit polynomial field; that field satisfies the criterion where the grid field does (the grid field’s discriminant is negative on the single band  $1.00R$  to  $1.81R$  with the same depth, the polynomial’s on wiggly sub-segments of it), but the polynomial’s first derivative is wrong by up to a factor two at the swirl maximum, and its normal-mode eigenvalues do not represent the flow’s: the columnar solver finds a fast near-front mode on the polynomial field and none on the grid column through the swirl maximum.

*The front column.* The second caution concerns the column: the one that carries the instability is the front’s, not the swirl maximum’s: on the grid column through the front at the height where the two-dimensional  $m = 1$  mode peaks, the columnar solver has an isolated unstable viscous  $m = 1$  eigenvalue with growth 0.81 in collapse units at  $kR = 28$  (converged in resolution and domain, 20 percent above the next eigenvalue, inviscid limit 0.83 to 0.89 and still rising slowly with  $k$ ), against 0.79 to 0.80 for the frozen-base two-dimensional rate at 20 percent of the remaining time. The columnar rate varies from 0.5 to 1.05 across the front’s thickness, the columnar eigenfunction

sits at  $1.3$  to  $1.4R$  where the two-dimensional mode peaks at  $1.0R$ , and on the inviscid x28 state, whose front is sharper and whose instability is the axial-shear (Kelvin-Helmholtz) mode of the free front, no column reproduces the two-dimensional rate at all (columnar rates  $0.4$  to  $0.7$  against  $0.92$ , at twice the radius); so the parallel-flow model is local, identifies the mechanism on the viscous front and brackets rather than predicts the two-dimensional rate; but it is a discrete, well-separated eigenvalue of a one-dimensional problem, which is the object a rigorous enclosure can reach (Section 7.3).

*Mechanism.* Its mechanism is centrifugal: on the front column the circulation  $r^2\Omega$  decreases outward beyond  $1.2R$ , so Rayleigh’s discriminant is negative there (on  $1.21R$  to  $2.0R$ ), and the inviscid  $m = 1$  rate approaches the local bound  $\sqrt{-\Phi_R} = 0.96$  collapse units from below as  $k$  grows ( $0.83, 0.87, 0.89$  at  $kR = 22, 44, 88$ ) with the eigenfunction narrowing onto the radius of the bound’s maximum, which is the fixed- $m$  large- $k$  limit of Billant and Gallaire [BG05, BG13]; the viscous peak at  $kR = 28$  is the Mayer-Powell cutoff of that branch [MP92]. The axial flow of the collapse extends the centrifugal instability to every azimuthal wavenumber [BG13], which is why the  $m = 0, 1$  and  $2$  short-wave families of Section 3.2 cross the threshold within a few percent of each other (on the front column the viscous columnar peaks are  $0.79, 0.81, 0.80$  and  $0.76$  collapse units for  $m = 0, 1, 2, 3$ , at  $kR = 22, 28, 33$  and  $44$  (the  $m = 3$  peak not yet passed), all in the same annulus: one centrifugal instability of the front at every azimuthal wavenumber, not three); without axial flow a Rayleigh-stable column is spectrally stable [GS20], and no rigorous or computer-assisted result exists for an unstable eigenvalue of a swirling column with axial flow.

Figure 5 collects the columnar spectra of the front column against wavenumber at both viscosities and inviscid, with the Rayleigh bound, the two-dimensional rate and the certified points.

*A certified discrete eigenvalue.* We therefore certify the discrete columnar problem. Let  $\Omega(s)$  and  $W(s)$  be the degree-30 polynomials in  $s = (r/r_{\max})^2$ ,  $r_{\max} = 2R$ , with exactly rational coefficients (the double-precision Chebyshev fit of the front column, converted exactly; the fit reproduces the grid column’s eigenvalue to  $0.01$  percent), and let  $(A_N, B_N)$  be the  $N$ -point Chebyshev-Gauss collocation pencil of the linearized viscous problem ( $m = 1, k = 2000, \nu = 5 \times 10^{-4}$ , no-slip imposed at the outermost collocation point  $r_{\max} \cos(\pi/4N)$ , axis conditions by parity) on the field  $(r\Omega, W)$ , with every entry enclosed in interval arithmetic. Then, by a Krawczyk verification with rigorous floating-point error bounds (Rump’s method; midpoint-radius matrices; all products bounded by the classical  $\gamma_n$  inequalities; the code and its tests accompany the paper), the pencil has a simple eigenvalue in the disc of radius  $1.2 \times 10^{-7}$  about  $\omega = -3.5051747 \times 10^4 + 5.0850344 \times 10^4 i$  for  $N = 150$ , with its eigenvector in an enclosed box in which the eigenpair is unique (the disc is not certified to contain no other eigenvalue; the nearest floating-point eigenvalue is  $1.05 \times 10^4$  away), and likewise for  $N = 100$  (radius  $5.2 \times 10^{-8}$ , agreeing to nine digits) and  $N = 60$  ( $5.0850368 \times 10^4$ , radius  $2.4 \times 10^{-8}$ ). An independent assembly of the pencil at 200 bits without the parity trick agrees with every stored entry within its radius, and a separate symbolic check reproduces the Cartesian linearized equations on a test field to  $1 \times 10^{-34}$ . The assumptions are those of every interval computation on BLAS and mpmath: the real matrix product is a classical inner-product algorithm, the interval library is correct, overflow is detected, underflow is floored. Toward the continuous operator, the defect of the certified eigenvector decays spectrally ( $2.4 \times 10^{-1}, 6.1 \times 10^{-3}, 5.2 \times 10^{-5}$  relative at  $N = 60, 100, 150$ ) and on the divergence-free subspace the operator is nearly normal at this eigenvalue (inverse-norm constant  $0.23$  against  $0.19$  for a normal operator, in units of  $1/(\nu k^2)$ , identical at the three resolutions); what remains is the rigorous bound on that constant. In collapse units the certified growth rate is  $0.812485 (T - t)^{-1}$  for each discrete problem. The same certificate at  $N = 100$  gives  $0.6580$  at Hou’s second-stage viscosity  $5 \times 10^{-3}$  ( $k = 1200$ ),  $0.8485$  for the inviscid problem ( $k = 2000$ ), and  $0.7886$  for  $m = 0$  ( $k = 1600, \nu = 5 \times 10^{-4}$ ), each with radius below  $1.3 \times 10^{-7}$ , so the discrete columnar instability is certified at both viscosities, inviscid, and

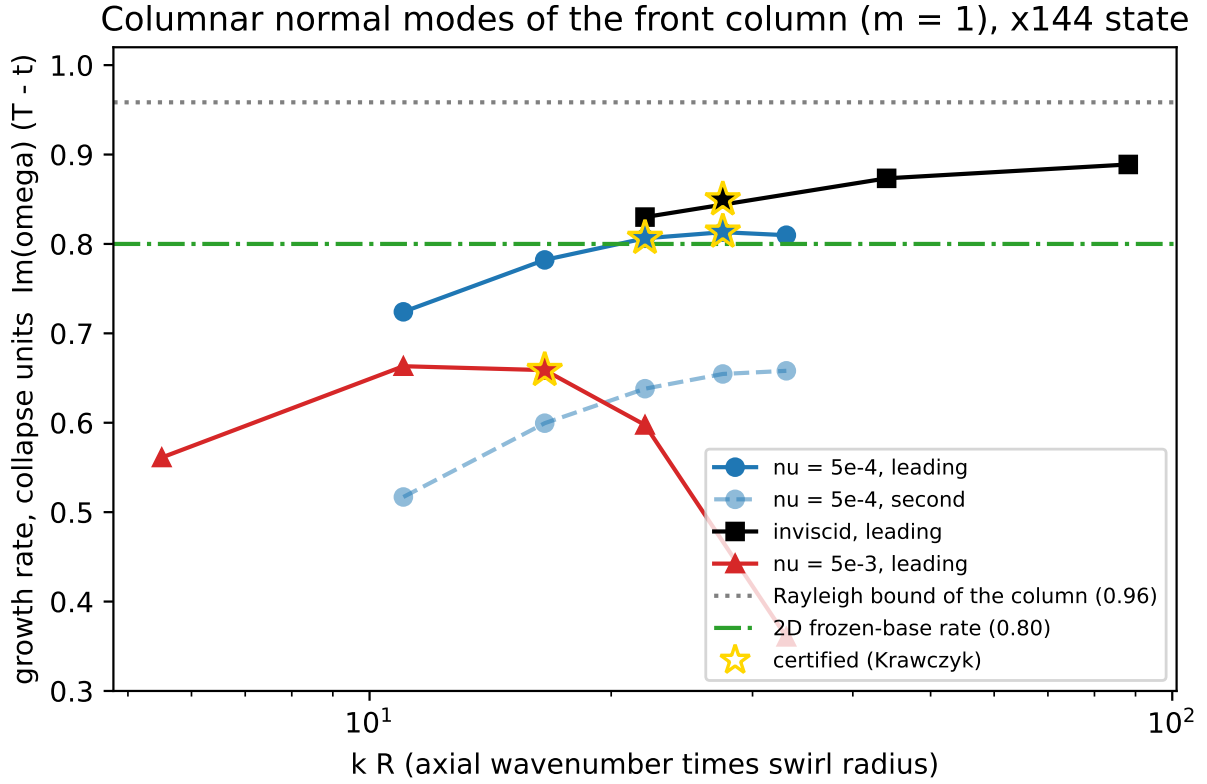


Figure 5: The columnar (parallel-flow) spectrum of the front column of the x144 viscous state versus axial wavenumber, in collapse units, at  $\nu = 5 \times 10^{-4}$  (leading and second eigenvalue), inviscid, and  $\nu = 5 \times 10^{-3}$ ; the Rayleigh bound of the column; the 2D frozen-base rate; the certified discrete eigenvalues (Krawczyk) at  $k = 1600, 2000$  ( $\nu = 5 \times 10^{-4}$ ), 1200 ( $5 \times 10^{-3}$ ), 2000 (inviscid).

at two azimuthal wavenumbers. This is a computer-assisted proof about the discretized problem on an explicit field; it certifies neither the continuous columnar operator (that needs defect and inverse-norm bounds of the Nakao-Plum type, and the nine-digit agreement across  $N$  is evidence for it, not proof) nor the two-dimensional collapse, whose operator has no isolated eigenvalue to enclose (Section 3.7).

### 3.7 The spectrum of the frozen operator

The rates of Sections 3.1-3.4 are growth rates of evolved perturbations, not eigenvalues. A direct eigenvalue computation of the frozen  $m = 1$  operator (Arnoldi iteration on the matrix-free linearization, largest real part) shows why the distinction matters. On the Euler x28 state at  $257 \times 512$  the leading eigenvalues are complex pairs with  $\sigma(T - t) = 3.14$  and  $3.04$  (frequencies  $\pm 115$  and  $\pm 110$  in collapse units), residual  $\|Lv - \lambda v\|/\|v\| = 7 \times 10^{-6}$ , with axial wavenumber centroid at half the resolved range and radial centroid just inside the swirl maximum: they are the short-wave shearing continuum of Section 7.1 at the grid scale, whose rate grows with resolution, and not the long-wave rate 0.3-0.5 that the localized seed excites. The long-wave direction is therefore not a leading discrete eigenvalue of the inviscid frozen problem, and an inviscid eigenvalue enclosure would be a statement about the grid.

The viscous operator ( $\nu = 5 \times 10^{-4}$ ) damps the short waves and should have a resolution-convergent leading eigenvalue; that is the object a rigorous enclosure would target. Plain Arnoldi did not reach it: at  $513 \times 1024$  the iteration failed to converge in 400 restarts, and at  $257 \times 512$  with a Krylov space of 60 it had not returned after 8.5 CPU-hours (about 30,000 operator applications) and was stopped. The leading eigenvalues sit in a cluster whose separation is of the order of the viscous damping of neighbouring short waves, which is the regime in which unshifted Arnoldi stalls [MR96, LS01, T11]; the remedies are a shift-invert or Cayley transform with an inner linear solve, or the exponential (time-stepper) transform of Edwards, Tuckerman, Friesner and Sorensen [ETFS94], Arnoldi on  $\exp(L\Delta)$ , which needs no inner solve and contracts the damped continuum to the origin. We tried both matrix-free routes: Arnoldi on the propagator over a tenth of the remaining time (190 applications, eight restart cycles) did not converge, and shift-invert with an inner GMRES solve stalled at a 36 percent inner residual for a real shift at half the threshold, with or without a shifted-diffusion preconditioner, which says the shifted operator is nearly singular over a region: the shift lies inside the pseudospectrum of the damped continuum rather than near an isolated eigenvalue. The leading viscous eigenvalue is therefore not isolated at this resolution, and this is the solver's view of the same continuum that Section 7.1 describes; it also bounds what a rigorous enclosure of the two-dimensional operator could achieve, since every enclosure radius scales with the inverse of the distance to that continuum. The reachable rigorous statement is the one-dimensional radial eigenvalue problem on the front column, certified at the discrete level in Section 3.6. The evolved-perturbation rates of Sections 3.2-3.3, certified by their resolution pairs, stand independently of this.

## 4 Nonlinear fate

The  $m = 1$  mode was seeded at 20 and 50 percent of the local swirl velocity on the Euler x28 state, at 20 percent on the viscous x144 state, and at 20 percent on the Euler x200 state, with  $m = 0, 1, 2$  (and 3) carried and the Reynolds-stress feedback on the base. Against the unperturbed collapse at matched times the axisymmetric amplitude lags: on Euler x28 by 0.65, 1.3, 2.1, 3 percent at 10, 13, 16, 21 percent of the remaining time with the 20 percent seed and by 5 percent at 11 percent

with the 50 percent seed, at which the axisymmetric swirl maximum stalls and falls while the three-dimensional swirl maximum stays flat at 1.9 times the axisymmetric one, the mode’s own growth drops below its linear rate (0.26 against 0.31), and the  $m = 2$  harmonic sits on the  $\epsilon^2$  curve at  $8 \times 10^{-3}$  of the  $m = 1$  energy without migrating. The x28 continuation reached 30 percent with a 17 percent lag and the three-dimensional swirl ratio flat at 1.37. On the viscous base the 20 percent localized seed lags by 0.2 percent at 9 percent and by about 7 percent at 24 percent.

A random seed on the viscous x144 base (three modes, seed 0.2, followed to 37 percent of the remaining time) tests whether the short-wave family rolls up into an intensifying substructure of the kind seen in strained mixing layers and tornado subvortices. The three-dimensional vorticity peak rose from 1.04 to 1.71 times the axisymmetric peak while the axisymmetric amplitude lagged Type I by 9 percent and the second harmonic stayed at two percent of the first. The end-state fields show the mode’s energy density peaking on the ring (radius 0.98 of the swirl maximum’s, height 0.91 of its) with half-widths of 0.7 to 0.8 collapse lengths, thirty times the Burgers thickness  $\sqrt{\nu(T-t)}$ , and the swirl at that point varying by a factor 1.38 around the azimuth. No compact off-axis maximum at the viscous scale, no Burgers-locked core, no climbing harmonics: the rising vorticity ratio is the linear family growing on a base it depletes.

The outcome is the generic saturating Reynolds-stress feedback of a supercritical mode (Stuart-Watson [SW60]), the mechanism by which Hou and Li’s simulation of Kerr’s antiparallel tubes [HL06, HL08] was depleted. A perturbation of relative size  $\epsilon$  reaches order one only after an amplification  $\epsilon^{-1/\lambda}$  with  $\lambda = \sigma - 1/2$  of order 0.1 to 0.3, which for  $\epsilon = 1 \times 10^{-2}$  is  $10^{10}$ : before the singular time in exact arithmetic, beyond any computation in practice, and depleting once it arrives. We found no seed, state or viscosity at which the symmetry-broken state continued to collapse.

## 5 The viscous continuation

### 5.1 The singular time and the amplitude exponent

Hou’s own caveat [H23b, Sec. 3.4] is that the Type I rate inequalities ‘would be almost impossible to verify numerically since it requires the exact value of  $T$ ’. We fit  $T$  by the log-derivative method: for  $A \sim C(T-t)^{-p}$  the ratio  $A/A'$  equals  $(T-t)/p$ , linear in  $t$  with slope  $-1/p$  and zero at  $T$ , so sliding-window fits give  $p$  and  $T$  together with nothing assumed, and Luo and Hou’s criterion [LH14a, LH14b] is that both stop moving as the window slides toward  $T$ . On the resolved snapshots of the continuation (Section 5.2) four-point windows ending at 0.002281, 0.002282, 0.002283, 0.002284 and 0.0022845 give  $p = 1.49, 1.37, 1.19, 1.04, 1.05$  with  $T = 0.0022888, 0.0022881, 0.0022872, 0.0022865, 0.0022866$ , and five-point windows ending at 0.0022845, 0.002285 and 0.0022853 give  $p = 1.04, 1.00, 1.06$  with  $T = 0.0022865, 0.0022864, 0.0022866$ : both have stopped moving over the last three windows, at  $p = 1.03 \pm 0.03$  and  $T = 0.0022865 \pm 1 \times 10^{-7}$ . The earlier apparent  $p \sim 1.8$  of a fixed- $T$  fit was the concave  $A/A'$  of subleading corrections in the early windows. A first reading of ‘falling Type I constants’ from a  $1/A$  fit with a drifting  $T$  was retracted on this basis.

### 5.2 The continuation and the front-following map

Hou’s first-stage solution was continued at  $\nu = 5 \times 10^{-4}$  past his switch time on a  $513 \times 1024$  grid with a moving radial map and an axial map concentrated at the symmetry plane, from 25 to 217 times the initial amplitude, with a twin at a different axial map and Courant number agreeing to 0.5 percent in amplitude through x217; past x217 the front’s width fell to two cells and the twins diverged by 27 percent, and the run was stopped. A 2048-point restart from x131 with a three-times finer axial map read the front at three cells at x268, still grid-limited. The solver was then given a

front-following axial map: a Gaussian point density centred on the vorticity maximum with a fifth of the points within three front widths, re-fitted with the radial rezone every hundred steps, the fields moved by exact sine-series evaluation; a map contrast above about 500 breaks the mapped Poisson solve, so the window is kept at three front widths (contrast about 140, symmetrization error  $2 \times 10^{-5}$ ), at which the restart reproduces the state to a percent.

On that map the same x222 state has a front  $4.8 \times 10^{-5}$  wide over ten cells where the coarse map had read  $7.7 \times 10^{-5}$  over three, and after one interval (x268) the front is  $3.3 \times 10^{-5}$  wide over nine cells. Re-evaluating the x268 state on a second map gives the error bars: front thickness  $\pm 15$  percent, vorticity plateau  $\pm 5$  percent, swirl constant  $\pm 2$  percent from the grid, meridional constant  $\pm 20$  percent, amplitude and its constants 0.5 percent.

With the window floor set at  $1.5 \times 10^{-4}$  (contrast 140 to 170) the continuation was carried on two grids (1536 and 2048 axial points) from 131 to 395 times the initial amplitude, agreeing to 0.5 percent, which is the range of every measurement in Sections 5.3 to 6. At about 400 times the initial amplitude ( $T - t$  about  $8 \times 10^{-7}$ ,  $\log(1/(T - t)) = 14.0$ ) the one-scale collapse ends: the primary front stalls (x395 to x382 over one output interval,  $A(T - t)$  falling from 4.3 to 3.7 with  $T = 0.0022866$ ) while a second structure nearer the symmetry plane, at  $z$  about 0.0003 against the front's 0.0009, grows to 96 percent of it; at the next interval the maximum is at the second structure (x484, at  $r = 0.00185$ ,  $z = 0.00030$ ), and at the next it has burst to x776 ( $r = 0.00158$ ,  $z = 0.00025$ ,  $A(T - t) = 5.6$ ). The two follower-map grids and a third run on a wide fixed fine region (2048 points, no following, contrast 175, which cannot lose the front) agree at every one of these points to 0.2 to 0.6 percent through x484 and to 3 percent at x776, so the transition is a property of the solution and not of the map. The aspect ratio  $z/r$  of the maximum's position drops from 0.42 to 0.16 across it, the swirl inside the maximum empties ( $|u_1|$  at 0.7 of the maximum's radius, on its row, falls from 15 percent of the maximum at x395 to 3 percent at x776, and at 1.05 of the radius from 83 to 28 percent, so the structure is a ring compact in  $r$ ), and the meridional velocity at the maximum turns from axial to radially inward ( $|u^r|/|u^z|$  from 0.05 to 20 to 67), which is the opening of the vacuum region between the front and the axis that Hou's Sec. 3.2.4 describes his data as avoiding. This has the sign of Hou's two-scale structure: in his Case 4 at the same viscosity (different data) a vacuum region for  $u_1$  opens between the front and the axis and the growth then slows under viscous dominance [H23b, Secs. 1.3, 3.2.4 and 3.6], and in his rescaled model at dimension three the one-scale profile drifts toward two scales, which he calls a mild scaling instability [H24, Secs. 1 and 5]; whether a vacuum region opens here, and whether  $u^r/u^z$  at the maximum follows the aspect ratio, is what the wide-map run's later points will show. Past the transition the collapse does not resume. The wide-map run and the two follower runs agree to one percent at every point through 909 times the initial amplitude, and the amplitude sequence 395, 382, 484, 776, 899, 909, 1025, 1422, 2501 carries  $A(T - t) = 4.3, 3.7, 4.1, 5.6, 5.4, 4.4, 3.7, 3.4, 3.0$  with  $T = 0.0022866$ : the constant of Section 5.3 is replaced by pulses of about 60 percent, local fits of the exponent give a singular time bouncing between 0.0022861 and 0.0022895 over consecutive intervals, and the front and the ring alternate as the site of the maximum. Within the range we can resolve, the state neither continues the one-scale collapse nor visibly arrests; no exponent is determined after 400 times the initial amplitude, and the last three points are on maps past their tested contrast. The measurements of Sections 5.3 to 6 are all made before the transition, and the range over which the one-scale law and the constants hold is 144 to 395 times the initial amplitude.

### 5.3 The constants

With  $T = 0.0022865$  the resolved snapshots from x131 to x331 (Figure effig:4) give  $A(T - t) = 3.85, 3.86, 4.00, 4.06, 4.10, 4.15, 4.16, 4.14$  (flat);  $R/\sqrt{T - t} = 2.38, 2.40, 2.34, 2.34, 2.33, 2.36, 2.29,$

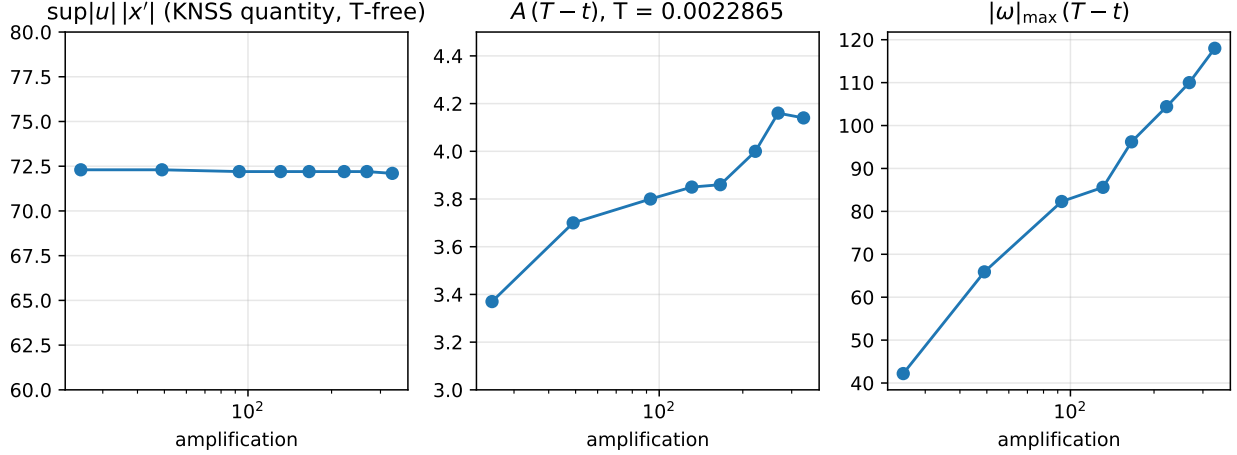


Figure 6: The constants against amplification (recorded values):  $\sup|u||x'|$  (the KNSS quantity,  $T$ -free),  $A(T-t)$  with  $T = 0.0022865$ , and  $|\omega|_{\max}(T-t)$ .

2.20;  $\sup|u|\sqrt{T-t}$  between 9.2 and 10.4, of which  $\pm 2$  percent is grid and the rest the singular-time sensitivity (a  $1 \times 10^{-7}$  change in  $T$  moves it 2 percent at  $T-t = 2 \times 10^{-6}$  and more later), so that its growth law is not readable from this data to the precision that would distinguish a logarithmic modulation from a small power or from a  $T$  error; the meridional Type I constant  $\sup|(u_r, u_z)|\sqrt{T-t} = 6.0$  to 7.7 (bounded);  $\omega(T-t) = 82, 86, 96, 104, 106, 110, 118$  (the x331 value from the run with the window floor; a second run whose window tightened without a floor degraded numerically at that point and is discarded), with its growth exponent in  $1/(T-t)$  falling from 0.43 (x131 to x222) to 0.33 (x268 to x331): levelling toward its Burgers ceiling;  $\omega/A = 21.4, 23.7, 25.5, 26.5, 28.5$  ( $T$ -free); the circulation maximum 64.9 to 64.7 (the maximum principle).

#### 5.4 The front is the Rott-Lundgren layer

The front's full width at half maximum is 0.86, 0.65, 0.62, 0.39, 0.29 collapse lengths  $\sqrt{T-t}$  and 3.9, 2.9, 2.8, 1.7, 1.3 Burgers lengths  $\sqrt{\nu(T-t)}$  from x131 to x268 (the first three grid-limited), and 1.0 at x331 (six cells). The compressive strain at the front, measured on the symmetry plane at the ring radius and on the axis, is 2.2 to 3.8 in units of  $1/(T-t)$ . A Gaussian fit of the azimuthal vorticity across the front at x268 (24 cells inside three standard deviations) gives a standard deviation of  $0.58\sqrt{\nu(T-t)}$ , a relative residual of 5 percent and a baseline of 9 percent of the peak; the strain implied by the thickness,  $\nu(T-t)$  over the variance, is 2.9 against the measured 2.2 to 3.3; the velocity jump implied by the fit,  $\sqrt{2\pi} \delta \omega_{\text{peak}} = 2600$ , agrees with the measured jump of  $u_r$  across the layer (2980) and with the integral of the vorticity (3240). The jump constant  $\Delta U \sqrt{T-t}$  is 2.8 to 3.3, a third of the swirl constant, and the sheet Reynolds number  $\Delta U / \sqrt{\nu a}$  is 120 to 150. At x331 (six cells) the fit residual is 10 percent and the implied strain 5.5 against a measured 3.8, and the  $u_r$  jump no longer accounts for the vorticity integral as the sheet tilts (the ratio of the radial to the azimuthal vorticity maximum rising from 1.1 to 2.1): the identification holds to the resolution available.

The front is therefore the exact strained-layer solution  $u = (-ax, v(x, t), az)$ ,  $v_t - axv_x = \nu v_{xx}$ , of Rott and Lundgren [R58, L82], under the collapse strain, with thickness  $\sqrt{\nu/a}$ . This is the configuration in which the peak vorticity scales exactly as  $1/(T-t)$  for a Type I velocity jump, and the vorticity's super- $1/(T-t)$  growth between x25 and x268 was its approach to this floor.

## 5.5 The local $L^3$ norm

The scale-invariant quantity, the  $L^3$  norm of the velocity in the ball of radius four collapse lengths about the axis point the ring converges to, was computed on the resolved snapshots from x4 to x268: 25.3, 25.0, 24.6, 24.1, 23.3, 23.2, 23.1, 23.0, 22.8, 22.6, 22.5, 22.4, 22.3, 22.0, a monotone decrease of 13 percent over four units of  $\log(1/(T-t))$ , with the balls of eight and sixteen collapse lengths decreasing faster. What this does and does not say is stated in Section 6.3.

# 6 The scenario against the theorems

## 6.0 The $T$ -free measurement that decides the class

The Liouville theorem of Koch, Nadirashvili, Seregin and Sverak [KNSS09] for bounded ancient axisymmetric solutions with swirl (their Theorem 5.3) has one hypothesis beyond boundedness:  $|v(x, t)| \leq C/|x'|$  for all  $t < 0$ , with  $|x'|$  the distance to the axis. If  $\sup_t \sup_x |u(x, t)| |x'|$  is finite up to  $T$ , the Leray rescalings about any axis point obey  $|u^{(\lambda)}(y, s)| \leq K/|y'|$  uniformly, so a Type I limit is a bounded ancient axisymmetric solution satisfying the hypothesis and is trivial, contradicting the normalization of the rescaling; and a point that is not Type I is handled directly by Theorem 6.1 of the same paper and by [CSTY08], which need no Type I assumption. Either way the scenario is regular; no bound on  $ru^r$  enters. The quantity to measure is therefore  $\sup |u| |x'|$  over the domain as a function of amplification, and unlike every velocity constant of Section 5.3 it does not involve the fitted singular time at all. On the resolved snapshots it reads 72.3, 72.4, 72.2, 72.3, 72.3, 72.3, 72.2, 72.2, 72.2, 72.2, 72.1 from 144 to 331 times the initial amplitude, constant to 0.3 percent while  $\log(1/(T-t))$  rises from 10.6 to 13.9; the value at the swirl maximum itself is 21 to 22; the Type I constant  $\sup |u| (|x| + \sqrt{T-t})$  is 75.4 to 74.6; and the circulation maximum is 64.9 to 64.7 by its maximum principle. Three things must be said about what this does and does not show. First, the logical status: the theorems (Chen, Strain, Tsai and Yau [CSTY08]; Koch, Nadirashvili, Seregin and Sverak [KNSS09], Theorem 6.1; Seregin and Sverak [SS09]) exclude a singularity if  $\sup |u| |x'|$  stays bounded up to  $T$  (the companion note states the proposition and its proof by citation, including the step from the periodic cylinder to the local theorems, which we flag for the analysts); a finite computation shows the bound over the resolved range, here 3.3 units of  $\log(1/(T-t))$ , and can establish membership in the excluded class only in that sense: what it shows is the absence, over that range, of the departure the scenario needs. Second, where the value comes from: the supremum 72 is attained in the outer flow (the circulation maximum sits at  $r$  about 0.40,  $z$  about 0.10, far from the core), while the same quantity at the swirl maximum is 21 to 22; so the bound the theorem needs is carried by the outer flow, and a logarithmic growth of the core quantity would not reach the supremum until it had grown by a factor of about three. The core quantity itself,  $|u|r$  at the swirl maximum, reads 21.1, 22.1, 21.8 at 144, 217 and 268 times the initial amplitude (20.5 at 331, where the map limits it), flat within its scatter, where a logarithmic growth over the range would read about 30 percent. Third, what Hou's own diagnostics do: the log-weighted circulation criteria of Lei and Zhang,  $\max |\log r|^{3/2} |\Gamma|$  and  $\max (\log r)^2 |\Gamma|$  over  $r < 0.1$ , grow steadily in our run (163 to 223 and 314 to 486 over the earlier part of the continuation), as they must for a core of constant circulation collapsing toward the axis, since the weight itself grows; they are not measurements of the hypothesis of any exclusion theorem. The quantity Hou names as the escape is the global supremum of  $|ru^r|$  in his generalized rescaled model [H24, Sec. 1.5 and 5.4.4], reported to grow like  $\log(1/(T-t))$  over amplifications from  $10^{15}$  to  $10^{30}$ ; his three-dimensional Navier-Stokes paper [H23b] reports no value of it, and our range of amplifications has no counterpart in his data, so the comparison is structurally unavailable here. For the record,  $|ru^r|$  restricted to three swirl radii

about the maximum is non-monotone over our range: 21.9 at the start, falling to 14.9 at 72 times the initial amplitude, rising to 21.7 at 266 times (a 45 percent rise over 1.3 units of  $\log(1/(T-t))$ , faster than logarithmic but from its minimum) and 19.1 at 329 times on the other map, with the two maps differing by 13 percent at the end;  $|ru^z|$  at the core rises slowly from 7.3 to 10.3; the global maxima (49 and 20) sit in the far field. A later pair of continuations at 1536 and 2048 axial points on the same map gives 15.2 to 18.0 over 263 to 395 times, the two grids agreeing to 3 percent, before the front-following map loses the front (the collapse region carries two maxima in  $z$  that a single-centre map cannot hold, which is what ended every continuation past about 330 times). Across the late-stage transition of Section 5.2 (x382 to x776, 0.34 units of  $\log(1/(T-t))$ , on the wide-map run) the core-restricted  $|ru^r|$  rises from 16.5 to 20.7 within three swirl radii and from 25.6 to 32.3 within  $r < 0.1$ ,  $|ru^z|$  at the core from 9.8 to 16.0, and  $|u|r$  at the swirl maximum from 20.9 to 24.1, while the global supremum  $\sup |u||x'|$  stays in the far field; so Hou's quantity moves in his direction at the regime change, by a quarter in a third of a log-unit, which is a jump between regimes and not yet a trend. On this quantity our data are therefore inconclusive: they show neither a sustained logarithmic growth nor its exclusion, and a comparison with Hou's observation would need a continuation many orders of magnitude longer on one map with the core window fixed in collapse units. The computed scenario therefore satisfies, as measured over the resolved range and independently of  $T$ , the hypothesis of the exclusion theorems, with the bound carried by the outer flow and the core value flat; whether the core quantities eventually grow is the one thing the data leave open.

## 6.1 The hypotheses of the exclusion theorems

Chen, Strain, Tsai and Yau [CSTY09, II] prove that axisymmetric strong solutions with swirl in  $\mathbb{R}^3$  satisfying  $|v| \leq C_*|t|^{-1/2}$ , with  $C_*$  arbitrary, are regular; no bound of the form  $|u| \leq C/r$  and no bound on  $ru^r$  appears among the hypotheses. Seregin and Sverak [SS09], Theorem 1.1, prove the local statement: for an axisymmetric weak solution in  $L_3$  with pressure in  $L_{3/2}$ , a Type I bound on the meridional velocity alone on a parabolic cylinder about an axis point gives regularity there; the hypothesis  $|v| \leq C/|x'|$  appears only in their Theorem 3.2; and their Lemma 3.3 places singular points on the axis. Hou's gloss that the theorems apply 'provided that  $|ru^r|$  and  $|ru^z|$  remain bounded' [H24, Sec. 1.5], and his identification of the logarithmic growth of  $|ru^r|$  and of the log-weighted circulation as the scenario's escape, therefore do not touch the hypotheses that are actually used; Barker [B25] makes the same observation in  $\mathbb{R}^3$  and flags the wall at  $r = 1$  as the point to check, which the locality of the Seregin-Sverak theorem [SS09] settles. The measured meridional constant of Section 5.3 is bounded. The escape the theorems leave is growth of  $\sup |u||x'|$  itself, or of the meridional Type I constant, not of  $ru^r$  (Section 6.0); for a velocity that is not asymptotically Type I the Liouville theorem for bounded ancient axisymmetric solutions with swirl is open [KNSS09, AB19] but Theorem 6.1 of [KNSS09] and [CSTY08] still apply under the  $|x'|$  bound; the sharpest log-supercritical criterion with swirl, Seregin's [S21], allows growth only like  $(\ln \ln 1/(T-t))^{1/224}$ . Our data cannot read the swirl constant's growth law to that precision (Section 5.3), so this escape is neither established nor excluded; every resolved observable, the amplitude exponent, the meridional constant, the front at the Burgers scale, is that of a Type I collapse.

## 6.2 The exact local solution and where finite energy acts

The layer of Section 5.4 is an exact Navier-Stokes solution for any strain history  $a(t)$  (Rott 1958 [R58]; the Lundgren transformation [L82] reduces it to the heat equation), and for  $a = 1/(T-t)$  its thickness is  $\sqrt{\nu(T-t)}$  exactly. It belongs to the stretched-vortex class of Gibbon, Fokas and

Doering [GFD99], in which finite-time singularities exist (Ohkitani-Gibbon; Constantin’s infinite family of exact Euler solutions of infinite energy [C00]; Maekawa, Miura and Prange’s backward self-similar Burgers-vortex blowup for Navier-Stokes with linear strain [MMP19]). These escape every theorem by having infinite energy: the KNSS Liouville theorem [KNSS09] needs bounded ancient solutions, and the layer rescales to a linearly growing one; the Seregin-Sverak argument [SS09] needs the rescaled limit to be a suitable weak solution; the Leray-profile theorems of Necas, Ruzicka and Sverak [NRS96] and of Tsai [T98] need a profile in  $L^3$  or with local energy control. Finite energy acts at a computable distance: a linear strain  $1/(T - t)$  gives velocity  $L/(T - t)$  at distance  $L$ , while a Type I bound limits it to  $M/\sqrt{T - t}$ , so the linear profile survives only to  $L \sim M\sqrt{T - t}$ , about four ring radii, inside the computational domain. The depletion is visible in the numbers: the free exact layer has circulation growing like  $T/(T - t)$  and a Type II velocity, ours has a Type I velocity and  $p = 1$ , so the layer’s circulation exponent is halved by the finite-energy feedback. A core-resolving computation can never falsify the local model, which is exact; it can falsify the global one, and what it shows is a Type I collapse with the front at the viscous scale and no departure to 331 times the initial amplitude.

### 6.3 What the local $L^3$ norm does and does not say

The decrease of Section 5.5 contradicts no theorem as stated. The global conditions (a singularity requires the whole-space  $L^3$  norm to blow up [ESS03, S12]) say nothing about shrinking balls; Barker and Prange’s logarithmic lower bound [BP21] concerns balls of radius  $(T - t)^{(1-\delta)/2}$  with a doubly exponentially small constant; the at-scale theorems (Barker-Prange 2020 [BP20], Theorem 2; Kang, Miura and Tsai [KMT21]) are lower bounds by a small universal constant, and the measured 22, which is  $4.4 \times 10^4$  in the dimensionless units of the theorems, satisfies them by a wide margin. A Type I profile predicts the scaled norm to approach a constant, and a mild decrease toward a plateau is consistent with that. The sentence the data supports is that the critical norm’s content at the collapse scale is not growing, the opposite of the trend a singularity drives into the global critical norm: evidence against, not proof.

## 7 Discussion

### 7.1 The continuum obstruction

Every rigorous construction of a log-corrected (Type II) singularity in a parabolic or dispersive problem follows one recipe: a stationary or self-similar profile of a limiting problem, its linearized spectrum with finitely many unstable directions and a resonant mode, and a modulation argument. For Navier-Stokes the limiting problem is Euler, viscosity being exactly marginal under the Leray scaling, and this framing exists only as an exclusion program (Seregin’s Type II papers, Barker and Prange [BP20, BP21]), never as a construction. Our table supplies the missing spectral ingredient for the leading profile and finds that it fails: the inviscid profile has no non-trivial long-wave unstable direction but a continuum of unstable short-wave directions at every azimuthal wavenumber, whose in-frame rate exceeds the collapse rate from 8 percent of the remaining time. This is the essential spectrum of Lifschitz and Hameiri [LH91], Friedlander and Vishik [FV91], and Vishik [V96] (whose radius equals the maximal Lyapunov exponent of the profile flow), which Chen and Hou reconcile with their stable boundary profile by working in a low-regularity weighted norm where the outgoing transport damps short waves, declining the high-regularity estimate whose coercivity they call unclear. A construction on the interior profile must either find such a norm or cut the continuum off with a viscosity large enough, which is what Hou’s tenfold viscosity does by hand

for the axisymmetric family and, by Section 3.4, cannot do for the non-axisymmetric one without stopping the collapse. The object that results is a viscously selected profile whose inviscid limit is not an attractor, and an exactly self-similar viscous profile is what the Leray-profile theorems forbid.

## 7.2 The Liouville problem and the log escape

Whether a conditional Liouville theorem could close the log escape was examined directly. The KNSS argument [KNSS09] uses  $|v| \leq C/|x'|$  at two lines, both as a statement uniform over rescalings, and a logarithmic weakening breaks both structurally for every exponent, not by a constant; the published growth-side results reach only a double logarithm to a small power (Pan’s exponent 0.028 [P16]; Seregin’s  $1/224$  in scaled norms [S21]), and the counterexample of Seregin, Silvestre, Sverak and Zlatos [SSSZ12] for drift-diffusion Liouville theorems sits at the next rung. The one hypothesis type that would enter the proof at the level of the source term, a one-sided monotonicity of  $\Gamma^2$  in the axial direction, is not satisfied by the computed profile: on the collapse side of the core the sign of  $d_z(\Gamma^2)$  is mixed (58 to 69 percent negative by weight). The escape reduces to a single unextracted number, the dependence of the axis Holder exponent in the critical-drift oscillation decay on the drift constant; polynomial dependence would close every logarithmic power below a threshold, exponential dependence none. We leave it as the one analytical question this scenario poses.

## 7.3 What remains open

Discretely self-similar and aperiodic (chaotic) Type I singularities are excluded only near the continuous case (Chae and Wolf [CW17a, CW17b]) and are equivalent, in the aperiodic case, to the Liouville problem for bounded ancient solutions, which is open in three dimensions; but a discretely self-similar singularity is Type I with a point singularity, and in the axisymmetric swirl class every Type I singularity is excluded, so this loophole lives in the non-axisymmetric class, where the relevant computation, a period- $\log(\lambda)$  or aperiodic orbit of the rescaled equations in three dimensions, has never been attempted for a blowup problem (the closest precedent is Gundlach’s computation of the Choptuik spacetime as a periodic orbit in similarity time [G95]). The octahedral Kida-Pelz class is unexcluded by any theorem but its Euler collapse was superseded by depletion at higher resolution, and no viscous arrest study of it exists.

On the rigorous side of our own results, Section 3.6 certifies the discrete columnar eigenvalue; the step from the discrete to the continuous columnar operator is the Nakao-Plum programme (a defect bound for the certified eigenfunction and a rigorous bound on the inverse of the shifted operator on its complement), carried out for Orr-Sommerfeld and Rayleigh-Benard problems at this size; the defect and the floating-point inverse-norm constant reported in Section 3.6 are favourable, the rigorous bound on that constant is the remaining work, and its completion would make the result the first rigorous unstable eigenvalue of a swirling column with axial flow. Neither step certifies the two-dimensional collapse, whose operator has no isolated eigenvalue to enclose (Section 3.7).

## 7.4 The regularity side

No finite list of computations certifies global regularity for all data, but certification for one datum is a finite computation in principle (Chernyshenko, Constantin, Robinson and Titi [CCRT07]; Dashti and Robinson [DR08]), discharged in practice only for mild data; for Hou’s datum the exponential tolerance collapses under a  $10^7$  amplification. What this work contributes to that side is characterization: a finite-codimension picture in which the collapse is Type I with a converged singular time,

the front sits at the viscous scale, the axisymmetric long-wave direction is stable, and the short-wave families, axisymmetric and not, outrun the collapse at both of Hou’s viscosities and deplete it when they arrive.

## 7.5 Predictions for a three-dimensional computation

No non-axisymmetric computation of Hou’s data exists. The first one will test the following statements, which are the content of this paper reduced to numbers that a single run at  $\nu = 5 \times 10^{-4}$ , started from the axisymmetric solution at 144 times the initial amplitude with a small random perturbation, can confirm or refute.

1. The azimuthal wavenumber  $m = 1$  energy in a thin band on the front grows, in the collapsing frame, faster than the collapse from 11 percent of the remaining time ( $\sigma(T - t) = 0.63$  at 14 percent and 0.77 at 18 percent against the threshold  $1/2$ ), with  $m = 0$  and  $m = 2$  following at 12 and 9 percent. The long-wave  $m = 1$  direction (a displacement of the collapse) stays below threshold throughout (0.44 at 51 percent).
2. The growth saturates as depletion, not as a sub-vortex: the three-dimensional vorticity maximum reaches 1.7 times the axisymmetric one while the axisymmetric amplitude falls 9 percent behind the Type I law (20 percent seed at 37 percent of the remaining time); the second harmonic stays at about two percent of the first; the mode’s energy peaks on the ring at 0.98 of the swirl maximum’s radius and 0.9 of its height with half-widths of 0.7 to 0.8 collapse lengths; no compact off-axis maximum forms at the viscous scale.
3. For a seed of relative size  $\epsilon$  the symmetry breaking becomes order one after an amplification of  $\epsilon^{-1/\lambda}$  with  $\lambda$  between 0.1 and 0.3 beyond the crossing, so a 1 percent seed reaches order one only after an amplification between  $2 \times 10^6$  and  $10^{20}$  of the amplitude at which it was introduced; the collapse does not continue through it.
4. At Hou’s second-stage viscosity  $5 \times 10^{-3}$  the same instability crosses at 27 percent of the remaining time (0.74 at 37 percent); the tenfold viscosity moves the crossing, it does not remove it.
5. Any resolved axisymmetric continuation of the  $\nu = 5 \times 10^{-4}$  solution gives  $\sup_x |u| |x'| = 72 \pm 1$  with no trend over the amplification range 144 to 331, the circulation maximum 64.7 to 64.9, the meridional Type I constant 6 to 8, an amplitude exponent  $1.03 \pm 0.03$  against a converged singular time  $0.0022865 \pm 1 \times 10^{-7}$ , a front of thickness 1.0 to  $1.7\sqrt{\nu(T - t)}$  fitting the Rott-Lundgren profile [R58, L82] to 5 percent, and  $\omega(T - t)$  rising through 118 with a decreasing exponent.
6. On one map with the core window fixed in collapse units, the core-restricted  $|ru^r|$  (within three swirl radii of the maximum) either stays within its present band of 15 to 22 or grows like  $\log(1/(T - t))$  beyond 331 times the initial amplitude; a resolved continuation to 1000 times adds 1.1 units of  $\log(1/(T - t))$  and decides between the two at the 10 percent level, which the present data cannot. (Partly tested after the prediction was written: over 382 to 776 times, across the late-stage transition of Section 5.2, the quantity rose from 16.5 to 20.7, above the band, in 0.34 log-units; whether that continues is the open part.)

A three-dimensional run in which the symmetry-broken state continues to collapse with its amplitude on the Type I law, or an axisymmetric continuation in which  $\sup |u| |x'|$  grows logarithmically

(about thirty percent over the range above), would refute the corresponding statement and with it the paper’s reading of the scenario.

## Data and code availability

All solvers, logs, snapshots at the reported times, the extraction scripts for the figures, and the dated record of results, corrections and retractions are in the repository accompanying this paper.

## References

- [AB19] D. Albritton, T. Barker, On local Type I singularities of the Navier-Stokes equations and Liouville theorems, *J. Math. Fluid Mech.* 21 (2019), art. 43. [arXiv:1811.00502]
- [B25] T. Barker, Quantitative classification of potential Navier-Stokes singularities beyond the blow-up time, arXiv:2510.20757 (2025).
- [BP20] T. Barker, C. Prange, Localized smoothing for the Navier-Stokes equations and concentration of critical norms near singularities, *Arch. Ration. Mech. Anal.* 236 (2020), 1487-1541. [arXiv:1812.09115]
- [BP21] T. Barker, C. Prange, Quantitative regularity for the Navier-Stokes equations via spatial concentration, *Comm. Math. Phys.* 385 (2021), 717-792. [arXiv:2003.06717]
- [BG05] P. Billant, F. Gallaire, Generalized Rayleigh criterion for non-axisymmetric centrifugal instabilities, *J. Fluid Mech.* 542 (2005), 365-379.
- [BG13] P. Billant, F. Gallaire, A unified criterion for the centrifugal instabilities of vortices and swirling jets, *J. Fluid Mech.* 734 (2013), 5-35.
- [BK96] K. N. Beronov, S. Kida, Linear two-dimensional stability of a Burgers vortex layer, *Phys. Fluids* 8 (1996), 1024-1035.
- [CH22] J. Chen, T. Y. Hou, On stability and instability of  $C^{1,\alpha}$  singular solutions to the 3D Euler and 2D Boussinesq equations, *Comm. Math. Phys.* 405 (2024), art. 112. [arXiv:2206.01296]
- [CH25] J. Chen, T. Y. Hou, Singularity formation in 3D Euler equations with smooth initial data and boundary, *Proc. Natl. Acad. Sci. USA* 122 (27) (2025), e2500940122; Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data I: analysis, arXiv:2210.07191; II: rigorous numerics, *Multiscale Model. Simul.* (2025), doi 10.1137/23M1580395. [arXiv:2305.05660]
- [CW17a] D. Chae, J. Wolf, Removing discretely self-similar singularities for the 3D Navier-Stokes equations, *Comm. Partial Differential Equations* 42 (2017), 1359-1374. [arXiv:1610.09464]
- [CW17b] D. Chae, J. Wolf, On the Liouville type theorems for self-similar solutions to the Navier-Stokes equations, *Arch. Ration. Mech. Anal.* 225 (2017), 549-572. [arXiv:1609.06962]
- [CCRT07] S. I. Chernyshenko, P. Constantin, J. C. Robinson, E. S. Titi, A posteriori regularity of the three-dimensional Navier-Stokes equations from numerical computations, *J. Math. Phys.* 48 (2007), 065204. [arXiv:math/0607181]

- [C00] P. Constantin, The Euler equations and nonlocal conservative Riccati equations, *Int. Math. Res. Not.* 2000 (9), 455-465. [arXiv:nlin/0004019]
- [CSTY08] C.-C. Chen, R. M. Strain, H.-T. Yau, T.-P. Tsai, Lower bounds on the blow-up rate of the axisymmetric Navier-Stokes equations, *Int. Math. Res. Not.* 2008, art. rnn016.
- [CSTY09] C.-C. Chen, R. M. Strain, T.-P. Tsai, H.-T. Yau, Lower bounds on the blow-up rate of the axisymmetric Navier-Stokes equations II, *Comm. Partial Differential Equations* 34 (2009), 203-232. [arXiv:0709.4230]
- [DR08] M. Dashti, J. C. Robinson, An a posteriori condition on the numerical approximations of the Navier-Stokes equations for the existence of a strong solution, *SIAM J. Numer. Anal.* 46 (2008), 3136-3150.
- [E21] T. M. Elgindi, Finite-time singularity formation for  $C^{1,\alpha}$  solutions to the incompressible Euler equations on  $\mathbb{R}^3$ , *Ann. of Math.* 194 (2021), 647-727. [arXiv:1904.04795]
- [ESS03] L. Escauriaza, G. Seregin, V. Sverak,  $L_{3,\infty}$ -solutions of the Navier-Stokes equations and backward uniqueness, *Russian Math. Surveys* 58 (2003), 211-250.
- [ETFS94] W. S. Edwards, L. S. Tuckerman, R. A. Friesner, D. C. Sorensen, Krylov methods for the incompressible Navier-Stokes equations, *J. Comput. Phys.* 110 (1994), 82-102.
- [FV91] S. Friedlander, M. M. Vishik, Instability criteria for the flow of an inviscid incompressible fluid, *Phys. Rev. Lett.* 66 (1991), 2204-2206.
- [GS20] T. Gallay, D. Smets, Spectral stability of inviscid columnar vortices, *Anal. PDE* 13 (2020), 1777-1832.
- [GFD99] J. D. Gibbon, A. S. Fokas, C. R. Doering, Dynamically stretched vortices as solutions of the 3D Navier-Stokes equations, *Physica D* 132 (1999), 497-510.
- [GHDG08] T. Grafke, H. Homann, J. Dreher, R. Grauer, Numerical simulations of possible finite time singularities in the incompressible Euler equations: comparison of numerical methods, *Physica D* 237 (2008), 1932-1936. [arXiv:0711.2284]
- [G95] C. Gundlach, The Choptuik spacetime as an eigenvalue problem, *Phys. Rev. Lett.* 75 (1995), 3214-3217. [arXiv:gr-qc/9507054]
- [H23a] T. Y. Hou, Potential singularity of the 3D Euler equations in the interior domain, *Found. Comput. Math.* 23 (2023), 2203-2249. [arXiv:2107.05870]
- [H23b] T. Y. Hou, Potentially singular behavior of the 3D Navier-Stokes equations, *Found. Comput. Math.* 23 (2023), 2251-2299. [arXiv:2107.06509]
- [H24] T. Y. Hou, Nearly self-similar blowup of generalized axisymmetric Navier-Stokes and Boussinesq equations, *arXiv:2405.10916* (2024).
- [HL06] T. Y. Hou, R. Li, Dynamic depletion of vortex stretching and non-blowup of the 3-D incompressible Euler equations, *J. Nonlinear Sci.* 16 (2006), 639-664.
- [HL08] T. Y. Hou, R. Li, Blowup or no blowup? The interplay between theory and numerics, *Physica D* 237 (2008), 1937-1944.

- [KMT21] K. Kang, H. Miura, T.-P. Tsai, An epsilon-regularity criterion and estimates of the regular set for Navier-Stokes flows in terms of initial data, *Pure Appl. Anal.* 3 (2021), 567-594. [arXiv:2006.13145]
- [KNSS09] G. Koch, N. Nadirashvili, G. Seregin, V. Sverak, Liouville theorems for the Navier-Stokes equations and applications, *Acta Math.* 203 (2009), 83-105. [arXiv:0709.3599]
- [LS01] R. B. Lehoucq, A. G. Salinger, Large-scale eigenvalue calculations for stability analysis of steady flows on massively parallel computers, *Int. J. Numer. Methods Fluids* 36 (2001), 309-327.
- [LS83] S. Leibovich, K. Stewartson, A sufficient condition for the instability of columnar vortices, *J. Fluid Mech.* 126 (1983), 335-356.
- [LH91] A. Lifschitz, E. Hameiri, Local stability conditions in fluid dynamics, *Phys. Fluids A* 3 (1991), 2644-2651.
- [LZZ17] Z. Lei, Q. S. Zhang, N. Zhao, Improved Liouville theorems for axially symmetric Navier-Stokes equations, *Sci. Sin. Math.* 47 (2017), 1183-1198 (Chinese edition; English text arXiv:1701.00868).
- [L82] T. S. Lundgren, Strained spiral vortex model for turbulent fine structure, *Phys. Fluids* 25 (1982), 2193-2203.
- [LH14a] G. Luo, T. Y. Hou, Potentially singular solutions of the 3D axisymmetric Euler equations, *Proc. Natl. Acad. Sci. USA* 111 (2014), 12968-12973.
- [LH14b] G. Luo, T. Y. Hou, Toward the finite-time blowup of the 3D axisymmetric Euler equations: a numerical investigation, *Multiscale Model. Simul.* 12 (2014), 1722-1776.
- [MMP19] Y. Maekawa, H. Miura, C. Prange, On stability of blow-up solutions of the Burgers vortex type for the Navier-Stokes equations with a linear strain, *J. Math. Fluid Mech.* 21 (2019), art. 46. [arXiv:1807.10341]
- [NRS96] J. Necas, M. Ruzicka, V. Sverak, On Leray's self-similar solutions of the Navier-Stokes equations, *Acta Math.* 176 (1996), 283-294.
- [MR96] K. Meerbergen, D. Roose, Matrix transformations for computing rightmost eigenvalues of large sparse non-symmetric eigenvalue problems, *IMA J. Numer. Anal.* 16 (1996), 297-346.
- [MP92] E. W. Mayer, K. G. Powell, Viscous and inviscid instabilities of a trailing vortex, *J. Fluid Mech.* 245 (1992), 91-114.
- [N84] J. C. Neu, The dynamics of stretched vortices, *J. Fluid Mech.* 143 (1984), 253-276.
- [P16] X. Pan, Regularity of solutions to axisymmetric Navier-Stokes equations with a slightly supercritical condition, *J. Differential Equations* 260 (2016), 8485-8529. [arXiv:1410.6260]
- [PV26] B. Pineau, V. Vicol, On rotated backwards self-similar solutions of the incompressible 3D Navier-Stokes equations, arXiv:2607.09619 (2026).
- [R58] N. Rott, On the viscous core of a line vortex, *Z. Angew. Math. Phys.* 9 (1958), 543-553.

- [S12] G. Seregin, A certain necessary condition of potential blow up for Navier-Stokes equations, *Comm. Math. Phys.* 312 (2012), 833-845. [arXiv:1104.3615]
- [S21] G. Seregin, A slightly supercritical condition of regularity of axisymmetric solutions to the Navier-Stokes equations, arXiv:2109.09344 (2021).
- [SS09] G. Seregin, V. Sverak, On Type I singularities of the local axi-symmetric solutions of the Navier-Stokes equations, *Comm. Partial Differential Equations* 34 (2009), 171-201. [arXiv:0804.1803]
- [SSSZ12] G. Seregin, L. Silvestre, V. Sverak, A. Zlatos, On divergence-free drifts, *J. Differential Equations* 252 (2012), 505-540. [arXiv:1010.6025]
- [SW60] J. T. Stuart, On the non-linear mechanics of wave disturbances in stable and unstable parallel flows. Part 1, *J. Fluid Mech.* 9 (1960), 353-370; J. Watson, Part 2, *J. Fluid Mech.* 9 (1960), 371-389.
- [SSF83] C. Sulem, P.-L. Sulem, H. Frisch, Tracing complex singularities with spectral methods, *J. Comput. Phys.* 50 (1983), 138-161.
- [T11] V. Theofilis, Global linear instability, *Annu. Rev. Fluid Mech.* 43 (2011), 319-352.
- [T16] T. Tao, Finite time blowup for an averaged three-dimensional Navier-Stokes equation, *J. Amer. Math. Soc.* 29 (2016), 601-674. [arXiv:1402.0290]
- [T98] T.-P. Tsai, On Leray's self-similar solutions of the Navier-Stokes equations satisfying local energy estimates, *Arch. Ration. Mech. Anal.* 143 (1998), 29-51.
- [VV20] A. Vasseur, M. Vishik, Blow-up solutions to 3D Euler are hydrodynamically unstable, *Comm. Math. Phys.* 378 (2020), 557-568. [arXiv:1908.05766]
- [V96] M. M. Vishik, Spectrum of small oscillations of an ideal fluid and Lyapunov exponents, *J. Math. Pures Appl.* 75 (1996), 531-558.
- [WLGB23] Y. Wang, C.-Y. Lai, J. Gomez-Serrano, T. Buckmaster, Asymptotic self-similar blow-up profile for three-dimensional axisymmetric Euler equations using neural networks, *Phys. Rev. Lett.* 130 (2023), 244002. [arXiv:2201.06780]
- [W25] Y. Wang, M. Bennani, J. Martens, S. Racaniere, S. Blackwell, A. Matthews, S. Nikolov, G. Cao-Labora, D. S. Park, M. Arjovsky, D. Worrall, C. Qin, F. Alet, B. Kozlovskii, N. Tomasev, A. Davies, P. Kohli, T. Buckmaster, B. Georgiev, J. Gomez-Serrano, R. Jiang, C.-Y. Lai, Discovery of unstable singularities, arXiv:2509.14185 (2025).

Every entry above was checked against the arXiv abstract page, Crossref, or the journal page in two independent passes; the *Found. Comput. Math.* volume and page numbers of [H23a] and [H23b] and the English-edition volume of [LZZ17] remain to be filled from the publishers' pages.